CSE221: Algorithms
Lab-02
Summer 2026

Action required for Students

[Google Form] CSE221 Lab Policy:
- The lab policy file can be found here - Link
- Please complete the lab policy acknowledgement form - Link

[Google Form] Codeforces Account Information: (Action required)
- Open an account at https://codeforces.com/
- Please submit your codeforces details here - Link

Students who have not completed this yet, please do so ASAP.
Otherwise, your lab quizzes and assignment submissions will not be graded.

Quiz 02

Date: July 04-05, 2026 (as per your lab class schedule)
Syllabus: Lab 02 content

Take Preparation for both assignment problems and outside assignment problems
[Student Drive Folder Link]

Assignment-02 Links:

codeforces Invitation Link:
https://codeforces.com/contestInvitation/712f07c3cdb839bd7349572942723bbbaca15376

Alternative Link: https://codeforces.com/contests/700634
[The alternative link will **only work** after you have successfully registered using the first link.]

Deadline: Jul 5, 2026 2:00 PM

Note: Use FAST I/O in your code to avoid Time Limit Exceeded (TLE), especially in the quiz.

After the deadline ends, you can submit your solution by entering the contest as a Virtual Participant for practice purposes. **However, please note, submissions after the deadline are not accepted for grading.**

Learning Outcome:

We have already learned about time complexity in theory. Now, we will focus on improving our naive solution approach. In this lab, we will explore the two-pointer technique. Some problems that can be solved using the two-pointer approach may also be solvable with searching techniques. However, our primary focus will be on solving them using the two-pointer method. Since searching is closely related to the two-pointer technique, this lab also includes a few problems on binary search.

Understanding Two-Pointer Technique

- Learn how to optimize naive solutions using the two-pointer approach.
- Understand when and why two-pointer techniques are applicable.
- Differentiate between the **opposite-direction** and **same-direction** two-pointer strategies.

Searching

- Review the concepts of **lower bound** and **upper bound** and their applications.
- Find out if there is any alternative way to solve the problem you have solved using two-pointer approach [Self Study]

Performance Analysis

- Compare brute-force, two-pointer, and binary search approaches in terms of time complexity.